# AAPL Historical Stock Analysis

*Insight Summary — Week 6 Advanced Python Analysis | AnalystLab Africa Internship, Batch B*

## 1. Dataset Overview

The analysis uses Apple Inc. (AAPL) daily OHLCV (Open, High, Low, Close, Adjusted Close, Volume) data spanning 5 years, from July 12, 2021 to July 9, 2026, pulled live via the yfinance API. The raw file required cleaning: datetime parsing, chronological sorting, duplicate-date removal, and forward/backward-fill for any missing values. The Date column was then set as the time-series index to enable resampling and rolling-window calculations.

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|---|---|---|
| Trading Days<br>**1,254** | Total Return<br>**+118.8%** | Avg. Daily Volume<br>**65.1M shares** |
| Price Range<br>**$125.02 - $316.22** | Positive Months<br>**56.7% (34/60)** | Best / Worst Day<br>**+15.33% / -9.25%** |

## 2. Data Transformation Techniques Applied

- Cleaning: datetime parsing, chronological sort, duplicate-date removal, and forward/backward-fill for missing OHLCV values.
- Calculated columns: Daily Price Change (Close − Open), Daily % Change, and High-Low Spread for every trading day.
- Filtering: isolated all days where Close exceeded $150, useful for studying the stock's behavior once it moved into a higher price regime.
- Grouping & aggregation: grouped Close by Year and Month to compute the monthly average closing price series.
- Resampling: month-end resampling of Close to compute month-over-month percentage returns.

## 3. Trends Identified

- The 7-day and 30-day moving averages show the stock moving through distinct multi-quarter regimes — a 2022 slide, a 2023-24 climb, a sharp 2025 air-pocket, and a 2026 push toward new highs — with MA crossovers roughly marking where each regime shifted.
- Rolling 30-day volatility is not constant: calm plateaus are interrupted by sharp spikes, confirming that risk clusters in time rather than staying flat.
- Trading volume tends to spike alongside the sharpest price moves, suggesting large moves are backed by above-average participation rather than occurring on thin volume.
- Monthly returns are positive in 34 of the last 60 months (56.7%), but individual down months are comparable in size to the best up months — a small number of extreme months disproportionately shape total return.

## 4. Features Created

- Daily Price Change and Daily % Change — the fundamental building blocks for all downstream time-series analysis.
- High-Low Spread — the intraday trading range, a simple proxy for daily volatility.
- 7-Day and 30-Day Moving Averages — trend-smoothing features used to identify regime shifts via crossovers.
- 30-Day Rolling Volatility — the rolling standard deviation of daily % change, used to quantify how risk varies over time.
- Monthly Average Closing Price and Monthly Returns — resampled series used to evaluate performance consistency month over month.

## 5. Key Insights

- Apple's total return over the period (+118.8%) looks like a smooth growth story only from a distance — the underlying daily and monthly data reveal a volatility story with a growth trend layered on top.
- A given trading day closes higher only about 53% of the time — closer to a coin flip with a slight positive lean than a guaranteed uptrend.

- The single best day (+15.33%) and single worst day (-9.25%) are outsized relative to typical daily moves, underscoring how much a handful of extreme days can shape long-run returns.
- Risk is not static: position-sizing or trading rules that ignore the rolling volatility feature would be poorly calibrated during high-volatility stretches and overly conservative during calm ones.

## 6. Recommendations

- Pair moving-average crossover signals with the rolling volatility feature to size positions dynamically — smaller size in high-volatility periods, larger size in calm ones.
- Investigate the largest single-day and single-month outliers against known events (earnings releases, macro announcements) to build an event-aware overlay in a future iteration.
- Backtest a simple moving-average crossover strategy against buy-and-hold to quantify whether it adds value net of added trading activity.
- Extend the pipeline to a peer stock or sector ETF for relative-strength comparison.

*Note: figures above are computed by the accompanying notebook (Advanced_Python_Analysis_AAPL.ipynb) using data pulled live via yfinance. Re-running the notebook will refresh these numbers as new trading days are added.*